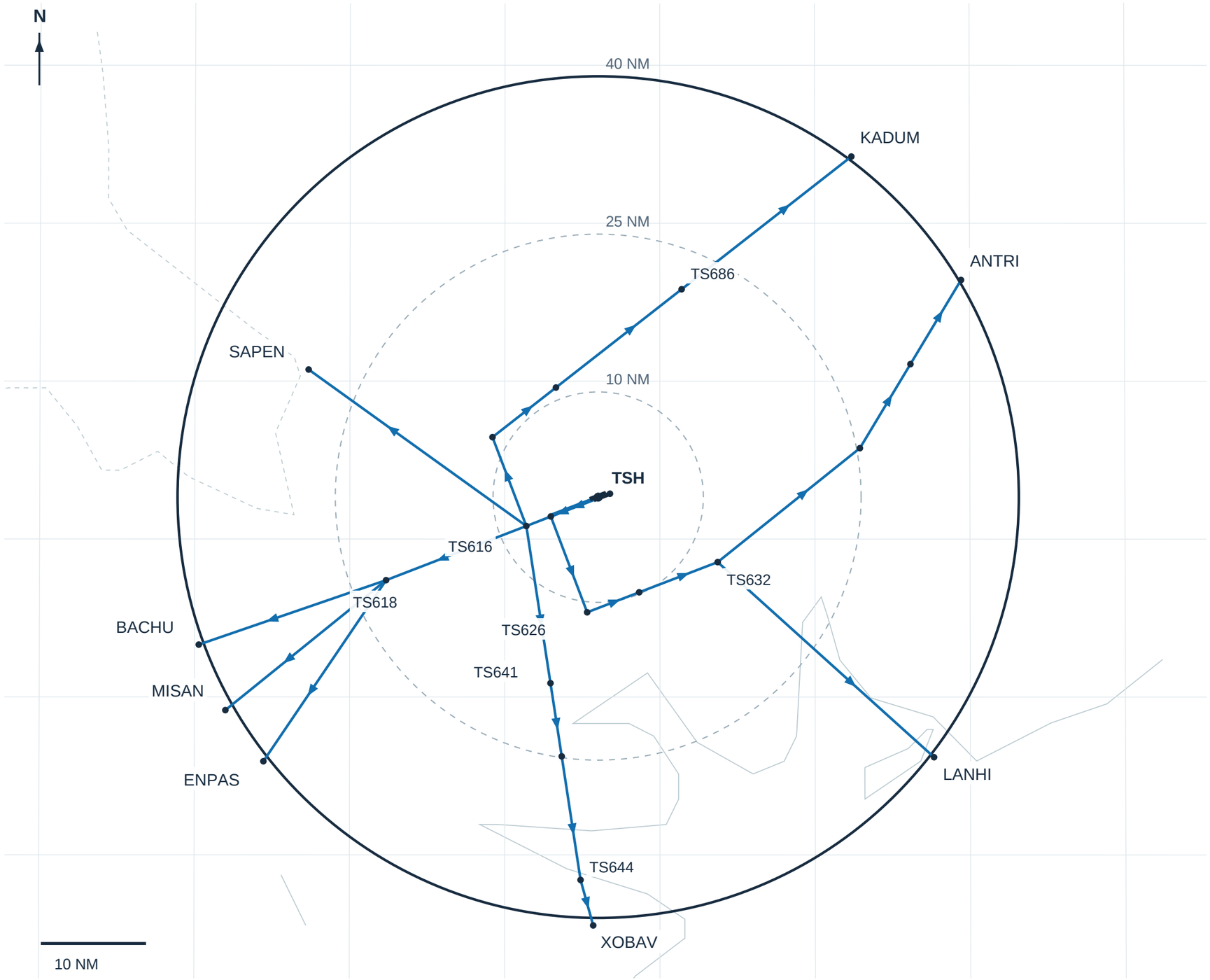
TSN TMA / 40 NM seminar study

Departures - runway 25L/25R concept, using supplied historical charts



North up. Spherical azimuthal equidistant projection centred on TSH.
Lines join chart fixes; fly-by turn arcs, protected areas and altitude constraints are not drawn.

SID sources: 83847 / 83854 (AMDT 03/24); conventional reference 10520 (30 Mar 2022). Coordinates: supplied WGS-84 workbook. Basemap: existing ASTRA context.

DESIGN BASIS

40 NM = 74.08 km. Centre proposed at TSH DVOR/DME: 10 49 06 N, 106 39 02 E.

Solid circle: assessment boundary. Dashed rings: 10 / 25 NM analysis bands, not ATS sector boundaries.

Published 2022 area chart: 75 km TSH radius annotation; 600-5050 m vertical label. Exact published outline is not digitised here.

FLOW INTERPRETATION

Blue: eight RNAV SID centrelines. Departures start westbound before branching northwest, southwest, south or northeast.

25L uses TS612 and 25R TS614 on the ANTRI / LANHI initial segment. Other depicted SIDs share TS616.

KADUM and ANTRI require coordination per their source charts.

Keep conventional SIDs as a separate scenario family after radial / DME reconstruction.

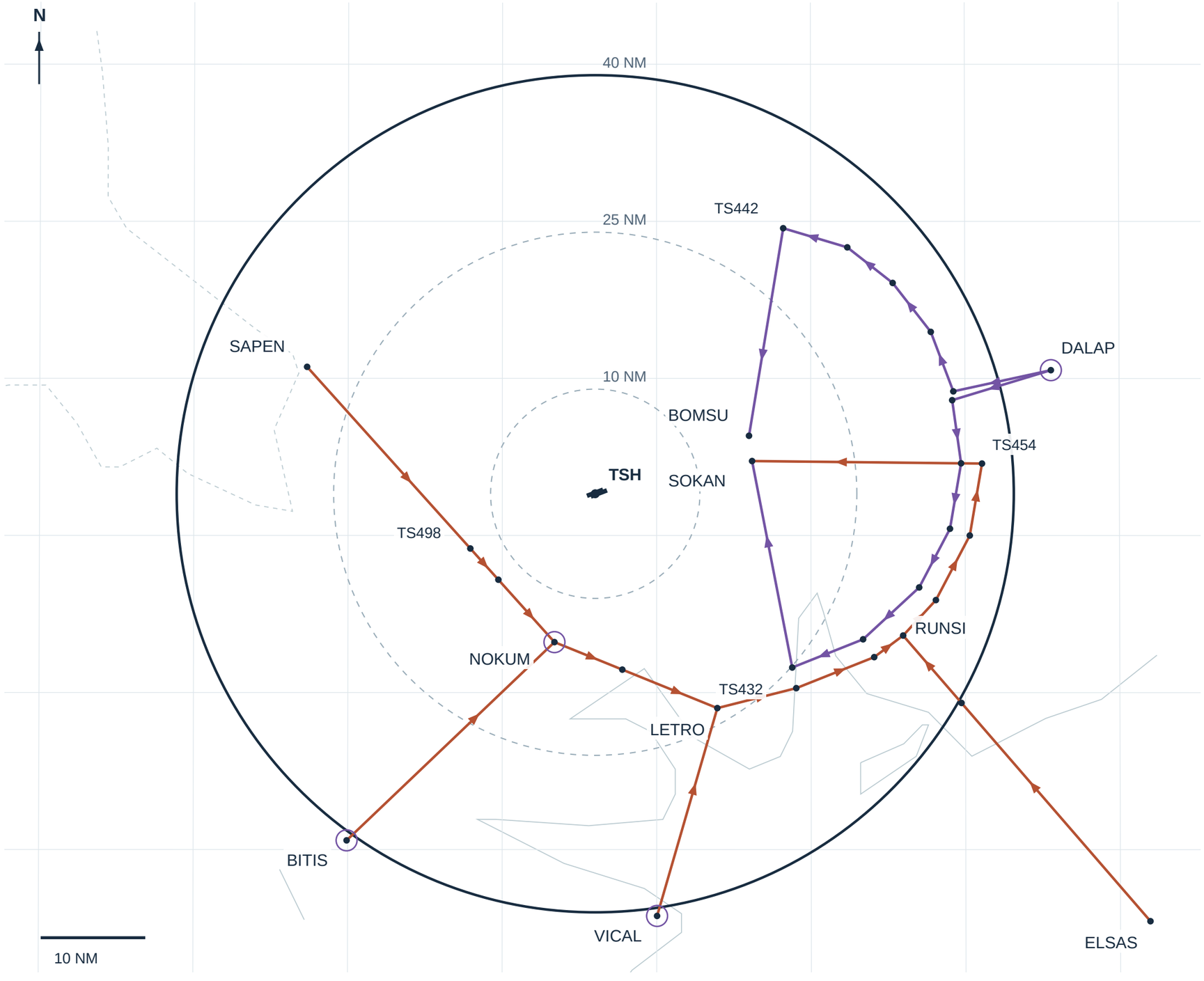
BOUNDARY TREATMENT

Retain external fixes and complete routes. DALAP is about 45 NM from TSH; ELSAS about 67 NM. Score the study area at 40 NM without moving fixes.

Academic concept draft. Historical source editions are not certified as one current operational dataset.

TSN TMA / 40 NM seminar study

Arrivals - runway 25L/25R concept, using supplied historical charts



North up. Spherical azimuthal equidistant projection centred on TSH.
Lines join chart fixes; fly-by turn arcs, protected areas and altitude constraints are not drawn.

STAR sources: 56080 / 56084 / 56090 (28 Dec 2023). Coordinates: supplied WGS-84 workbook. Basemap: existing ASTRA context.

DESIGN BASIS

40 NM = 74.08 km. Centre proposed at TSH DVOR/DME: 10 49 06 N, 106 39 02 E.

Solid circle: assessment boundary. Dashed rings: 10 / 25 NM analysis bands, not ATS sector boundaries.

Published 2022 area chart: 75 km TSH radius annotation; 600-5050 m vertical label. Exact published outline is not digitised here.

FLOW INTERPRETATION

Rust: SAPEN, BITIS, VICAL and ELSAS streams feed the southern/eastern path to SOKAN.

Purple: DALAP 2H routes south to SOKAN; DALAP 2J routes north to BOMSU.

Small purple rings mark holding fixes, not holding-pattern geometry.

Evaluate NOKUM / LETRO / RUNSI merges and SOKAN handoff demand. Geometric merges alone do not prove conflicts.

BOUNDARY TREATMENT

Retain external fixes and complete routes. DALAP is about 45 NM from TSH; ELSAS about 67 NM. Score the study area at 40 NM without moving fixes.

Academic concept draft. Historical source editions are not certified as one current operational dataset.